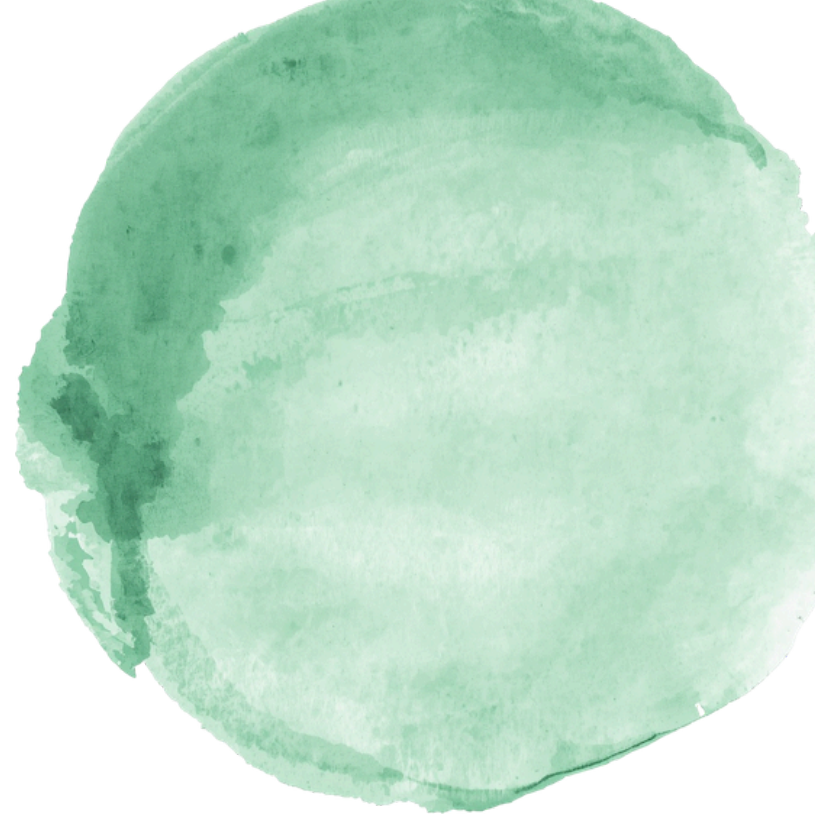




RESPIRATION – YEAST BREATHES TOO!





Does Yeast Breathe? How Do We Know?

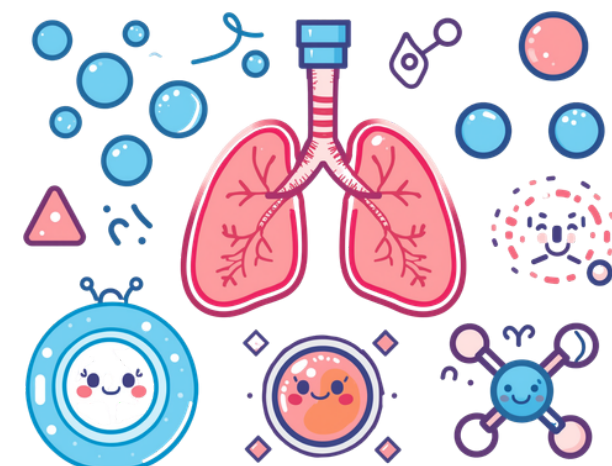
- Today we will discover something amazing! Yeast is a tiny thing that comes in packets. But is it alive? Can it breathe like us?
- Let's find out together with a fun experiment! Get ready to become little scientists and solve this mystery!



What is Breathing?



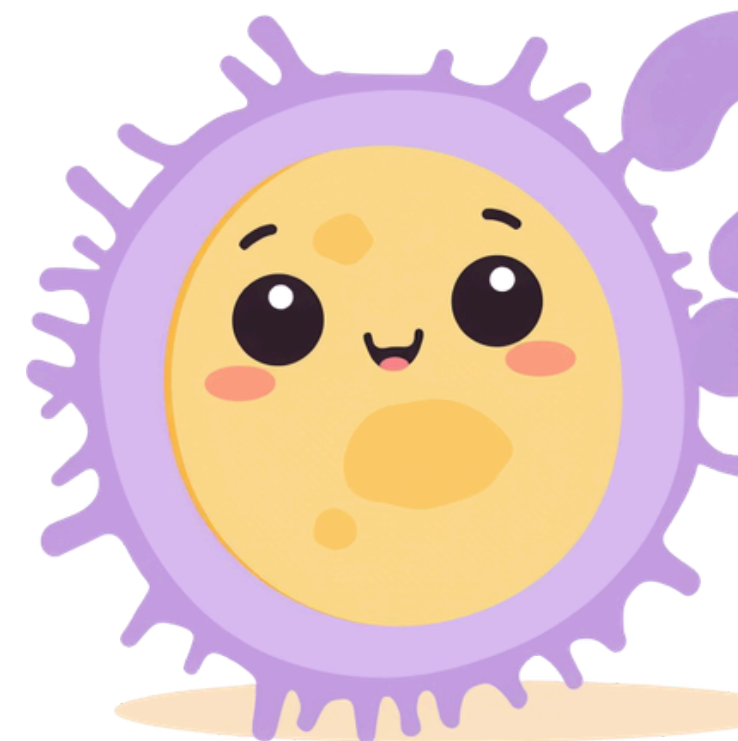
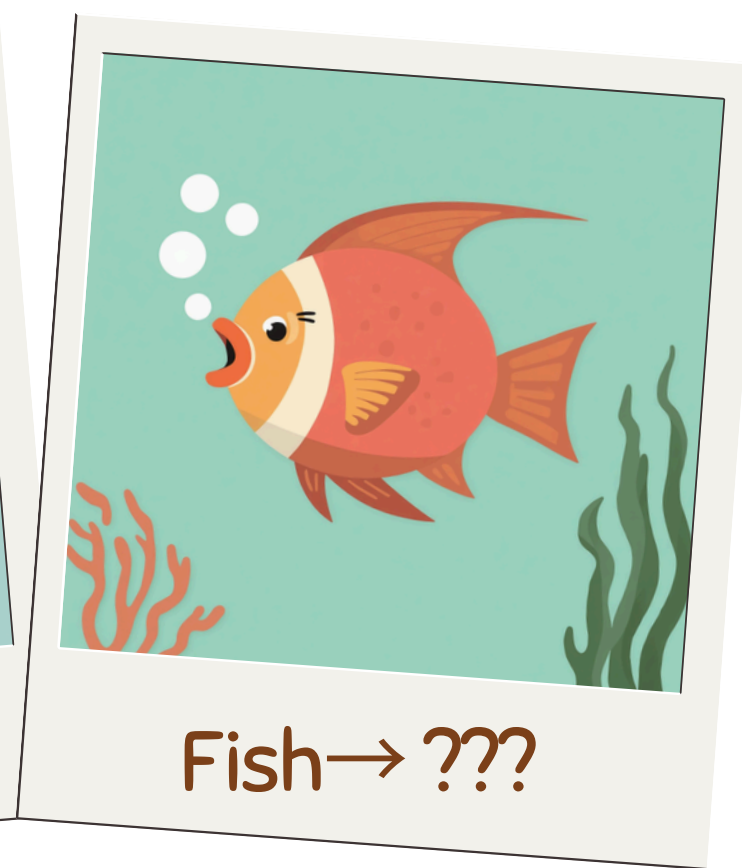
- All Living Things Breathe!
- We breathe to stay alive and have energy. Every living thing needs to breathe — from tiny plants to big animals!
- When we breathe, we take in air and use it to make energy. This helps us run, play, think, and grow!





Who Else Breathes?

Human → Breathes! Dog → Breathes! Fish → Breathes!



Experiment Steps 1-2



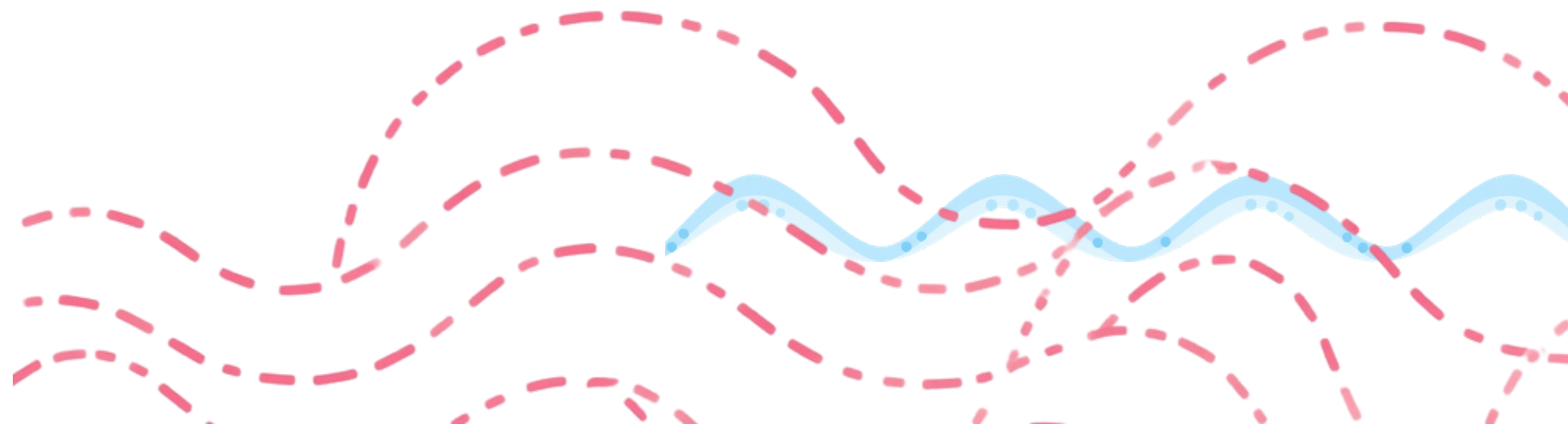
Step 1

Pour warm water into the bottle.
Make sure the water is warm, not
hot!



Step 2

Add yeast into the bottle. Watch
the tiny yeast grains fall into the
water!



Experiment Steps 3-4

Step 3

Add sugar to the bottle — this is yeast's food! Sugar gives yeast energy to breathe.



Step 4

Stretch a balloon over the bottle opening. Make sure it fits tightly so no air escapes!





Step 5: Wait and Watch!



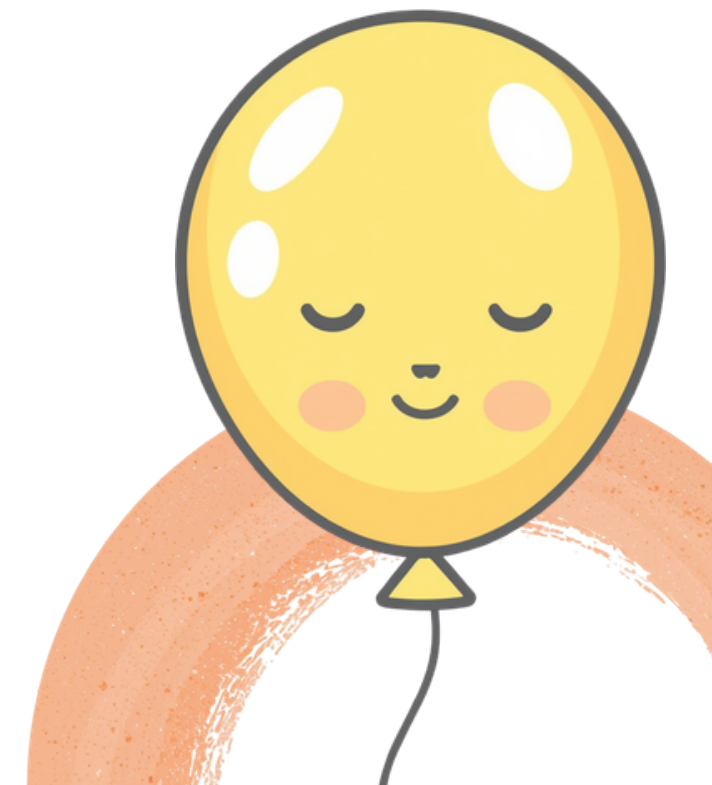
- Now it's time to observe! Look closely at your bottle and balloon. What changes do you notice happening?
- What do you see? What do you smell? Watch carefully and share your observations with your friends!

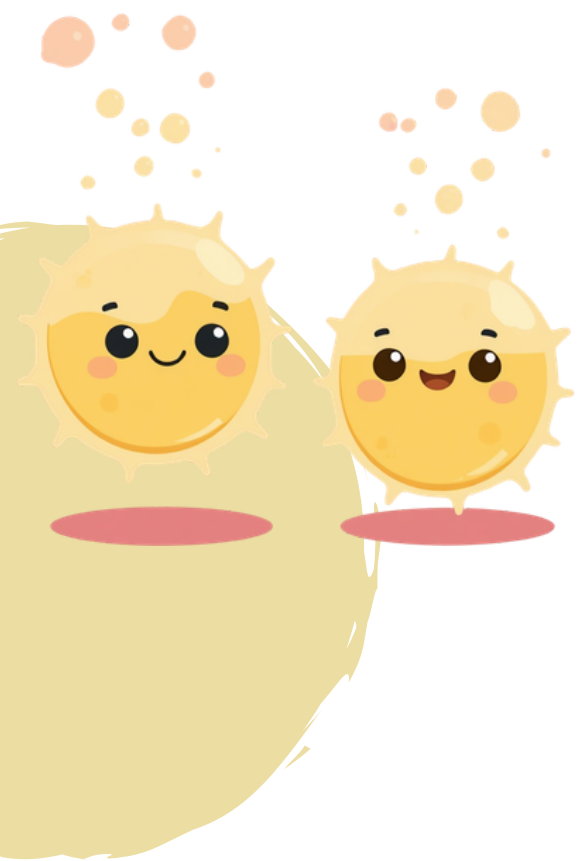


What we discovered in the process



The Balloon Grew! Yeast ate the sugar and breathed. It made bubbles — carbon dioxide gas! The gas filled up the balloon. That's how we know yeast is alive and breathing!





Yeast is Alive!



- Yeast is a tiny living thing — so small you cannot see it with your eyes! But it is alive, just like you and me.
- Yeast breathes just like us! When it breathes, it takes in food (sugar) and releases gas. That gas makes bubbles!
- The bubbles yeast makes are called carbon dioxide. This is the same gas we breathe out! Yeast is a living, breathing thing.
- Even though yeast is so tiny, it does big things! It helps make our food fluffy and delicious by breathing and making bubbles.



Yeast cell

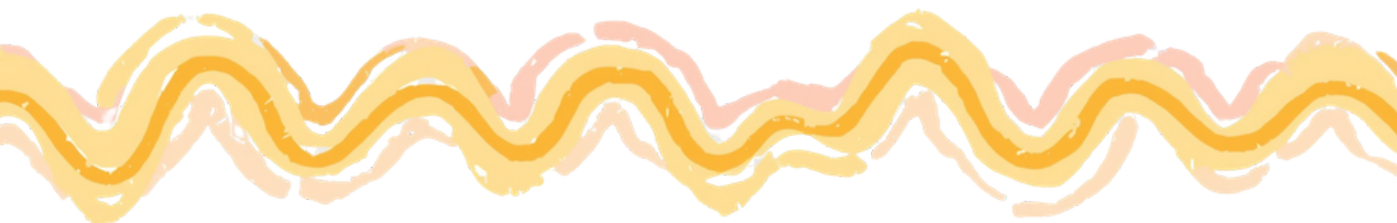


Indian Kitchen Science

– Idli & Dosa



- Yeast's Cousins Make Our Food Fluffy!
- Idli batter bubbles overnight. That's tiny living things breathing!
- Just like yeast, tiny living things in idli batter breathe and make bubbles. That's why our idlis are so soft and fluffy!





Indian Kitchen Science – Bread

Bakery Uncle Uses Yeast! Yeast breathes in the dough, makes gas bubbles, our pav becomes soft and fluffy!



What We Learned!

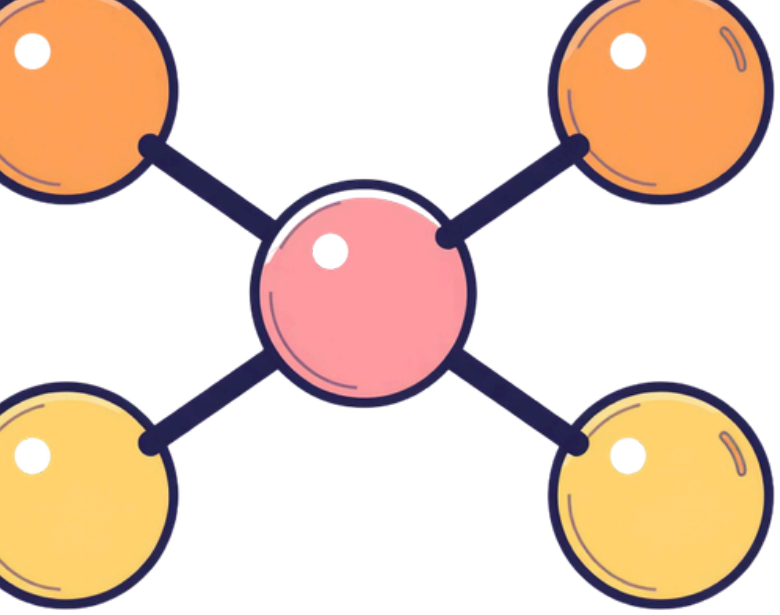


- ✓ Yeast is ALIVE
- ✓ Yeast BREATHES
- ✓ Yeast makes BUBBLES (gas)
- ✓ That's why idli and bread are FLUFFY!

Try This at Home!

Watch idli batter bubble tonight! Ask your family to show you the batter before and after it rises. You're seeing tiny living things breathe!





Thank You!

Keep exploring science at home!

